

CVE Intelligence Assistant Report

CSAI 810 - Project Report
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1. Project Overview

The CVE Intelligence Assistant is a Retrieval-Augmented Generation (RAG) system for cybersecurity vulnerability analysis. It enables users to query and understand CVEs (Common Vulnerabilities and Exposures) using natural language. The system targets response latency <5s, answer accuracy >80%, and operates within an academic budget.

2. Dataset Summary

Dataset	Coverage	Freshness	Role
NVD 2020–2024	130K+ CVEs	2–6 hr lag	Primary layer — semantic search baseline
CISA KEV	1,100 CVEs	Weekly	Confirmed exploits — urgency escalation
Exploit-DB	15K+ CVEs	24–48 hr	Weaponization proof via exploit code

3. Models and Tools Used

Component	Choice	Details
LLM	Llama 3.1-70B (Groq)	86% accuracy, 1.8s latency, \$47/mo — best balance of accuracy, speed & cost
Embedding	Sentence Transformers	Semantic vector search over CVE descriptions
Vector DB	FAISS / ChromaDB	Fast similarity retrieval for RAG pipeline
Framework	LangChain + Python	RAG pipeline orchestration and API integration
Data Sources	NVD API, CISA KEV JSON, Exploit-DB CSV	Public domain; fetched via REST API and bulk download

4. Steps to Run the Demo

1. **Install dependencies:** pip install langchain groq faiss-cpu sentence-transformers requests
2. **Set API key:** export GROQ_API_KEY=your_key_here
3. **Ingest data:** Run python ingest.py—it fetches NVD (2020–2024), CISA KEV, and Exploit-DB and builds a vector index.
4. **Launch assistant:** Running python app.py starts the query interface (CLI or Gradio UI).
5. **Ask a question:** e.g., "What is CVE-2021-44228?" The system retrieves context and returns a structured answer.

5. The source code & Video link

Drive link:

https://drive.google.com/drive/folders/1ZfO1a6qmMgY46CIeb0WSkQlHu1aJstlr?usp=drive_link

The source code: <https://huggingface.co/spaces/yomnafarag95/omago-cve-assistant>